



MANAGERIAL ACCOUNTING (135)

—Post-secondary—

STATE – 2019

Multiple Choice & Short-Answer Section:

Multiple Choice (15 @ 2 points each) _____ (30 points)

Short-Answers (12 @ 4 points each) _____ (48 points)

Production Portion:

Job 1: Budgeting Scenarios _____ (100 points)

Job 2: Variance Analysis _____ (96 points)

TOTAL POINTS _____ ***(274 points)***

**Graders: Please double check and verify all scores and
answer keys!**

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Workplace Skills Assessment Program competition.



General Instructions

You have been hired as a Financial Assistant and will be keeping the accounting records for Professional Business Associates, located at 5454 Cleveland Avenue, Columbus, Ohio 43231. Professional Business Associates provides accounting and other financial services for clients. You will complete jobs for Professional Business Associates' own accounting records, as well as for clients.

Unless otherwise instructed, round all answers to 2 decimal places.



Multiple Choice

- 1) B
- 2) D
- 3) D
- 4) D
- 5) A
- 6) B
- 7) D
- 8) C
- 9) A
- 10) B
- 11) C
- 12) D
- 13) A
- 14) A
- 15) C



Short-Answer (4 points each)

Star Labs maintenance cost over the last six (6) months is below. Use the high-low method to estimate the fixed and variable components of the maintenance cost.

Month	Maintenance Hours	Maintenance Cost
July	18	\$6,919
August	12	5,134
September	15	6,007
October	21	6,951
November	24	8,267
December	6	3,470

1) Variable Cost per unit **\$266.50**

2) Fixed Cost in total **\$1,871**

3) If maintenance hours for the following January are expected to be 22, estimate the total maintenance cost for the month.

\$7,734

Vibe manufactures sunglasses that sell for \$180 per pair and whose variable costs are \$135 per pair. The annual fixed costs are \$575,000. Use this information to compute the following:

4) Contribution margin per unit **\$45**

5) Contribution margin ratio **25% OR 0.25**

6) Break-even point in units **12,778 (GRADER: must be a whole number)**

7) Break-even point in dollars of sales **\$2,300,000**



Vibe adds a second version of sunglasses that sell for \$220 per and whose variable costs are \$155 per pair. Vibe expects the ratio to be 3:1 version 1 to version 2. Fixed costs are now \$625,000.

8) What is the weighted-average contribution margin per unit? **\$50.00**

9) How many of the Version 1 sunglasses will be sold at break-even? **9,375**

10) How many of the Version 2 sunglasses will be sold at break-even? **3,125**

Based on plans to produce 24,000 units, Vibe budgeted fixed costs to be \$300,000 and variable costs to be \$264,000. If actual production was 20,000, what would the flexible budget report show for:

11) Fixed Costs? **\$300,000**

12) Total Variable Costs? **\$220,000**



Problem 1: Budgeting Scenarios (100 points)

Grader: Each answer worth 5 points

Near the end of 2018, Star Labs prepared the following estimated balance sheet for 2018.

Star Labs					
Estimated Balance Sheet					
December 31, 2018					
ASSETS			LIABILITIES & EQUITY		
Cash	\$431,000		Accounts Payable	\$65,000	
Accounts Receivable	105,000		Taxes Payable	10,000	
Inventory	175,000		Notes Payable (due in 2022)	390,000	
Total Current Assets		\$711,000	Total Liabilities		\$465,000
Equipment (net)		472,500	Common Stock	472,000	
Total Assets		\$1,183,500	Retained Earnings	246,000	
			Total Stockholders' Equity		718,500
			Total Liabilities & Equity		\$1,183,500

Star Labs currently sells only one product, which is purchased for \$35 per unit and resold for \$60. Management expects to sell 7,000 units in the first quarter, with sales increasing by 10% each month. Calculate the Sales Budget for each month in the first quarter.

Sales Budget (in dollars)	January	February	March
	\$420,000	\$462,000	\$508,200

Management would like to have ending inventory equal to 20% of next month's sales. Calculate the Merchandise Purchases Budgets for each month in the first quarter.

Merchandise Purchases Budget (in dollars)	January	February	March
	\$123,900	\$274,890	\$302,400

Sales Commissions equal to 20% of sales are paid each month. General and administrative expenses are \$144,000 per year and are paid evenly throughout the year. Maintenance expense is \$2,000 per month and Depreciation Expense is \$1,200 per month. Calculate the total amount of General, Selling, and Administrative Expenses for each month in the first quarter.

General, Selling, & Administrative Expenses	January	February	March
	\$99,200	\$107,600	\$116,840



Management estimates that 30% of all sales are cash sales, and the remaining 70% of sales are on account. Of the credit sales, approximately 60% are collected in the month of sale, with the other 40% collected in the month after sale. Determine the expected cash collections for each month in the first quarter.

Cash Collections	January	February	March
	\$407,400	\$450,240	\$495,264

Historically, management has paid for 20% of merchandise purchases in the month of purchase, and the remainder in the month following purchase. Salaries total \$204,000 a year, and all employees are paid monthly. Determine the expected cash payments for the first quarter.

Cash Payments	January	February	March
	\$187,780	\$260,498	\$396,032

Determine the total cash budget for the first quarter.

Cash Budget	January	February	March
	\$650,620	\$840,362	\$939,594

Determine the balances in the following accounts at the end of the first quarter.

Accounts Receivable	\$142,296
Accounts Payable	\$241,920



Problem 2: Variance Analysis (96 points)

Grader: Each answer worth 8 points – 5 points for amount & 3 points for Favorable (F) or Unfavorable (U)

West-Allen Manufacturing Corporation manufactures headphones. During the last quarter of 2018, 315,000 headphones were manufactured and the following cost data was accumulated.

Standard Cost Information		
	Quantity	Cost
Direct Materials	2 parts	\$0.45 per part
Direct Labor	0.02 hours	\$8.00 per hour
Variable Manufacturing Overhead (Allocated based on Direct Labor Hours)	0.02 hours	\$10.00 per hour
Fixed Manufacturing Overhead (\$85,500 for static budget volume of 285,000 units and 5,700 hours, or \$15 per hour)		
Actual Cost Information		
Direct Materials	627,000 parts	\$332,310
Direct Labor	5,250 hours	\$43,050
Variable Manufacturing Overhead		\$56,700
Fixed Manufacturing Overhead		\$78,000

Compute the following variances and indicate whether each variance is Favorable (F) or Unfavorable (U)

Direct Materials Price Variance	\$50,160 U	Variable Overhead Cost Variance	\$4,200 U
Direct Materials Usage Variance	\$1,350 F	Variable Overhead Efficiency Variance	\$10,500 F
Total Materials Variance	\$48,810 U	Total Variable Overhead Variance	\$6,300 F
Direct Labor Rate Variance	\$1,050 U	Fixed Overhead Cost Variance	\$7,500 F
Direct Labor Efficiency Variance	\$8,400 F	Fixed Overhead Volume Variance	\$6,750 U
Total Direct Labor Variance	\$7,350 F	Total Fixed Overhead Variance	\$750 F